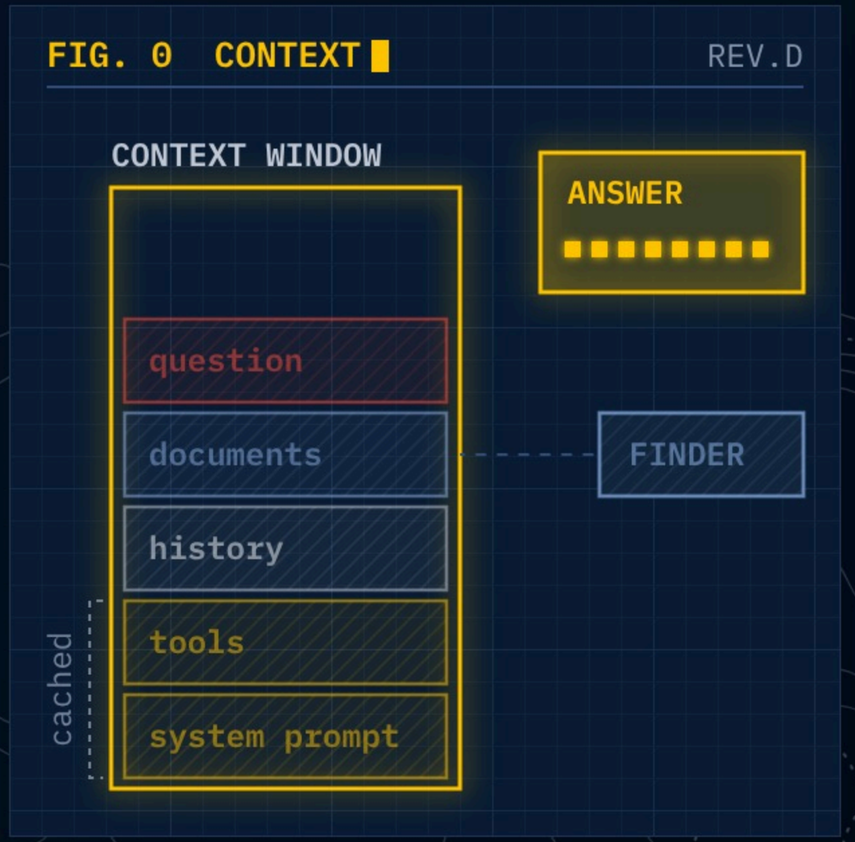
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TRAIN-THE-TRAINER SERIES · SESSION 4 OF 8

Tokens, context and how AI finds your data

The model only sees what you send it and what the finders return.



7–minute video | Stages 1–2 | Guide 2.2 · 2.3

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TODAY

By the end, your learners can…

1. Explain what a token is and why it matters
2. Read what fills a context window
3. Estimate the token cost of a workflow
4. Pick the right finder for your data

CHAPTERS · 7 MIN

drawn to scale



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WHY THIS MATTERS

Three complaints, one root cause: what the model can see

01

“The bill surprised us”

Cost comes from context re-sent in loops and long chats, not from the answer.

02

“It forgot what I said”

Long chats fill the window; early turns get dropped or lose attention.

03

“It couldn’t find the table”

The wrong finder was used: chunking flattened the table into prose.

The question for this session:

What does the model see, and what does it cost?

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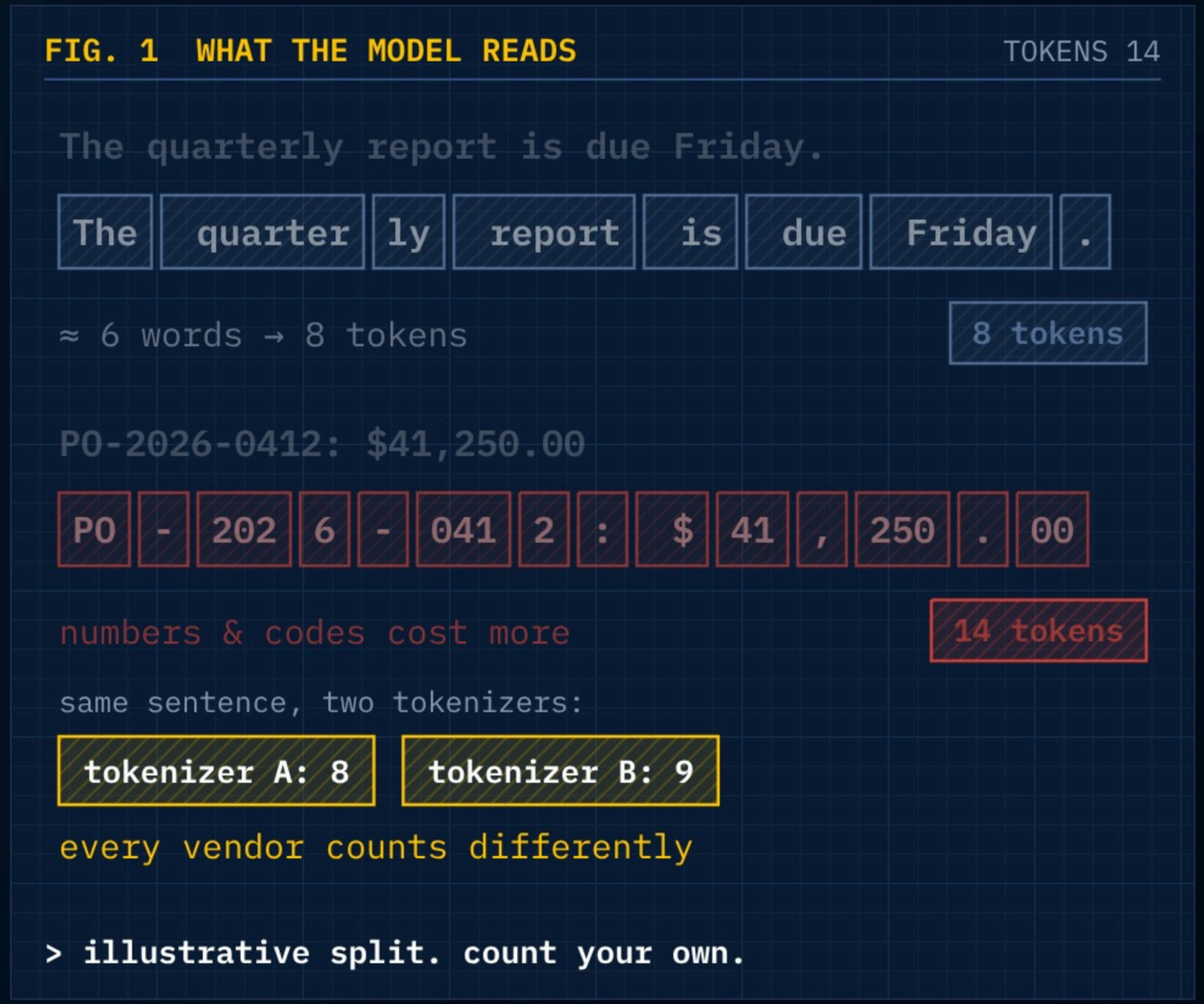
TOKENS

A token is the piece of text the model reads and writes

About ¾ of a word
Roughly four characters of English text

Codes and numbers cost more
Code, tables, IDs and other languages split into more tokens

Every vendor counts differently
Each model has its own tokenizer, so counts are estimates across vendors



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TYPES OF TOKENS

Not all tokens are the same, or cost the same

INPUT

Input tokens

Everything sent: instructions, tools, documents, history, question

OUTPUT

Output tokens

The answer; priced several times higher, and the main driver of wait time

THINKING

Thinking tokens

The model's working before it answers; billed as output

CACHED

Cached tokens

A stable prefix stored by the provider and re-read at a discount

IMAGE · AUDIO

Image and audio

Pages, pictures and recordings converted into tokens

TOOL

Tool tokens

Every connected tool adds its description to each request

EMBEDDING

Embedding tokens

Charged when documents are indexed for search

Rule

What you send costs more than what you get back

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THE CONTEXT WINDOW

One request, one fixed budget for input and output

Everything competes

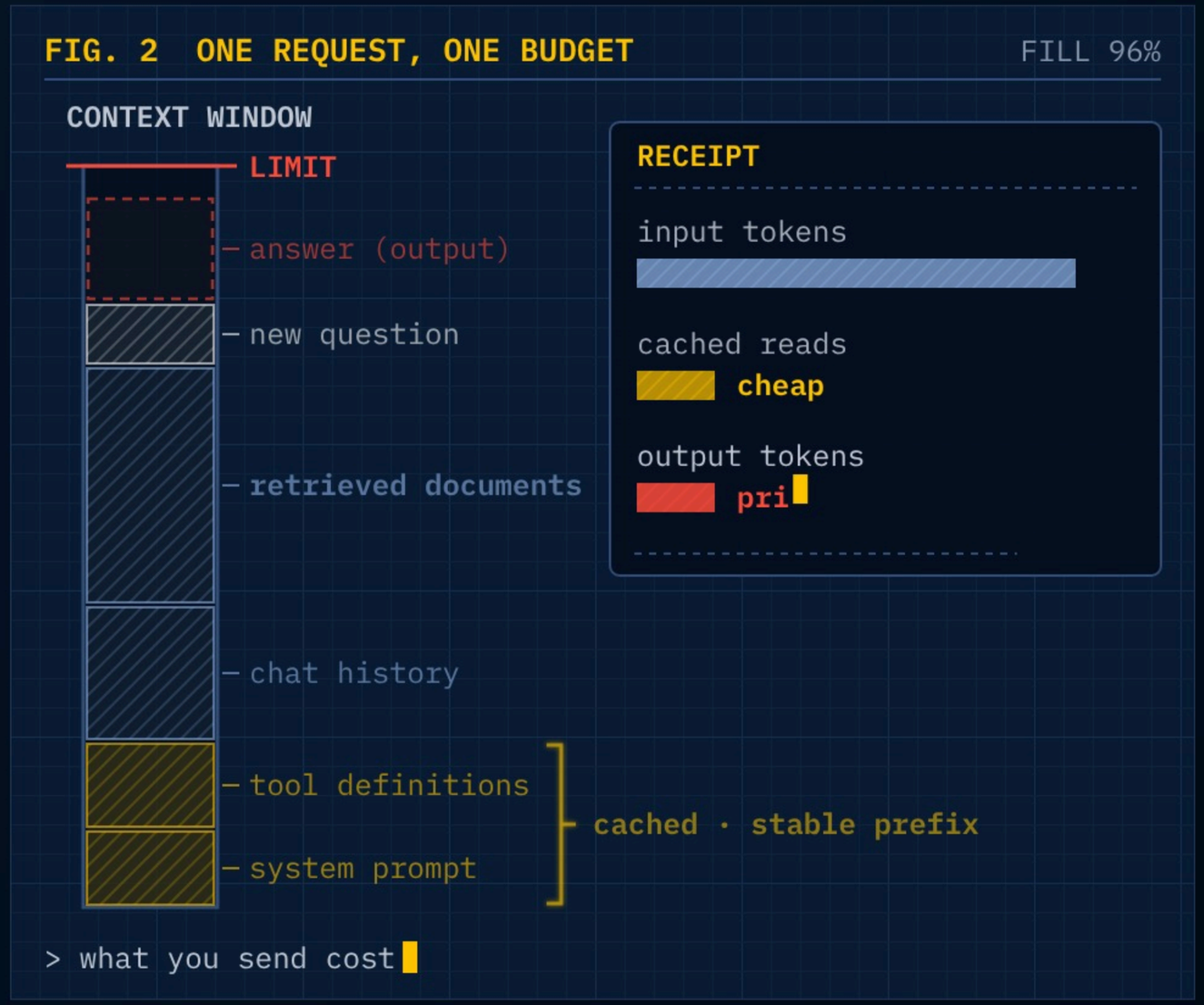
Instructions, tools, history, documents and the answer share one window

Cache what's stable

A repeated prefix is stored and re-read cheaply

Input dominates

What you send costs more than what you get back



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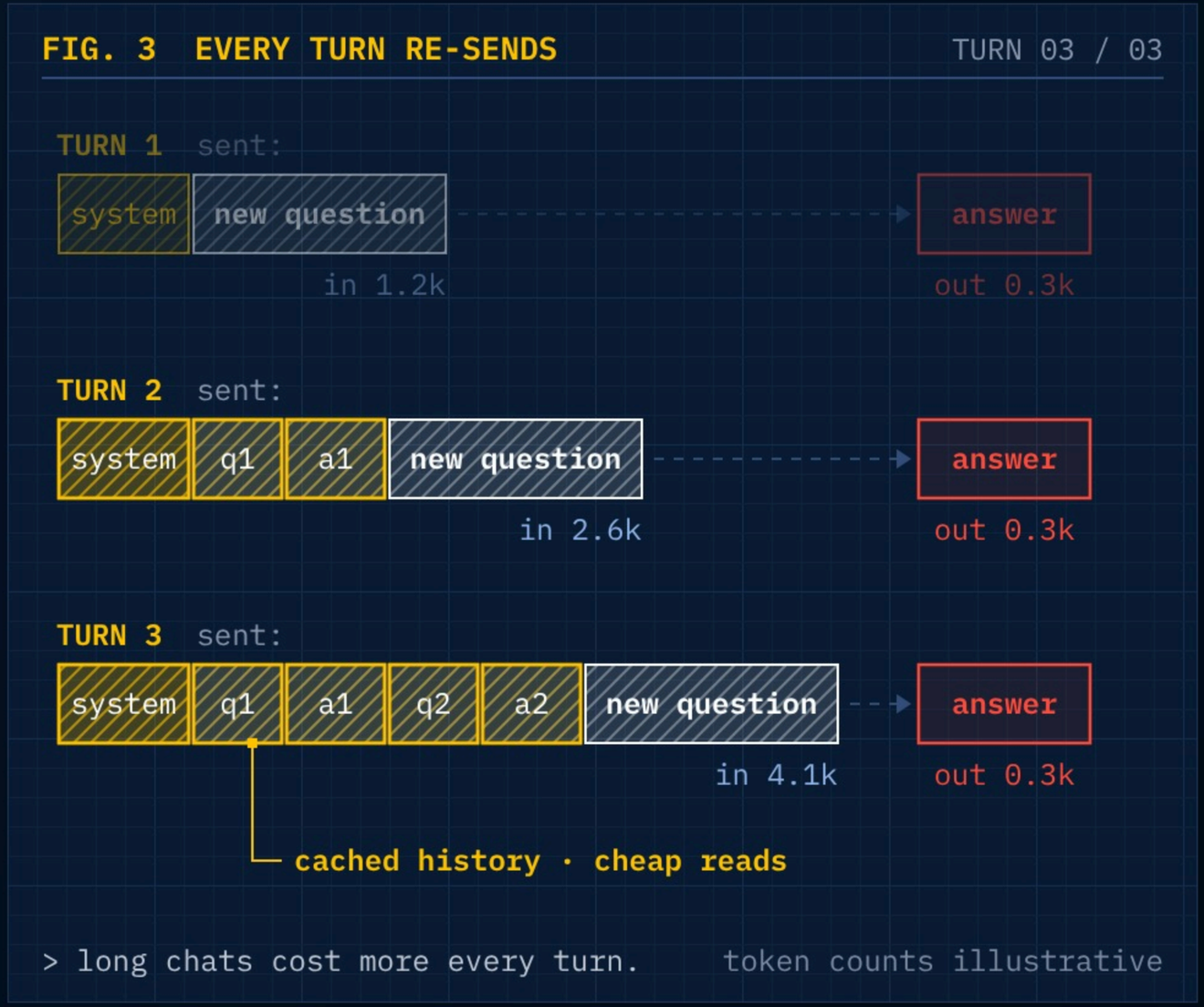
TOKENS IN A CHAT

Every turn re-sends the whole conversation

The API is stateless
Each turn sends the system prompt and every earlier message again

Input grows, output doesn't
Long chats cost more every turn

Caching softens it
The repeated history becomes cheap reads



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WHY TOKENS MATTER

Six reasons tokens show up in your work

01

Cost

Tokens per run × runs per month

02

Capacity

One window shared by everything you send

03

Speed

Output tokens are written one at a time

04

Rate limits

Tokens per minute are capped; batch jobs get throttled

05

Quality

A lean, organized context beats a stuffed window

06

Swapping models

Counts, costs and limits change with the tokenizer

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HOW AI FINDS YOUR DATA

The model only sees what the finders hand it

Finders, not the model

The harness runs the searches; the model reads the results

Several at once

Different data types need different finders

Merged and cited

Results are combined and handed over with citations



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SIX WAYS AI FETCHES DATA

No single finder covers every kind of data

01

Everything in context

Small, stable sets; cost grows with every question

02

Semantic search

Finds by meaning in large prose; weak on exact IDs and tables

03

Keyword search

Exact names, IDs and codes; weak on synonyms

04

Agentic search

Plans its own searches over live files; costly on huge sets

05

Structured queries

Exact records from tables and databases; needs an approved connection

06

Multimodal

Scans, slides and charts in their layout; newest and costlier

SECURITY An index is a copy of your data. It needs the same access rules and approvals as the source.

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MATCH THE FINDER TO THE DATA

Each kind of data has a finder that fits it

DATA TYPE	FINDERS THAT FIT	WHAT TO WATCH
Text documents and memos	Semantic + keyword search together	Cite the page, not just the file
Tables and spreadsheets	Structured queries	Totals come from a query, not from the model reading rows
Images, scans, slides	Multimodal retrieval or a vision model	Check numbers read off charts and scans
Code and configuration	Agentic search	Indexes go stale; read the live files
Live records	Structured queries only, through an approved connection	Access control on the query; log every lookup

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THE RULE OF THUMB

Send less.
Cache what's stable.
Fetch with the finder that fits.

The model only sees what you send it and what the finders return.

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EXERCISE

Count it, then find it

Pause & try

A Count it

1. Paste one non-sensitive prompt into an approved token counter.
2. Note input and expected output tokens.
3. Tokens per run × runs per month.
4. What would a shorter prompt or caching save?

B Find it · which finder?

1 What's our policy on remote work?	**2** Find ticket INC-20418
3 Total spend by vendor this year	**4** What does the org chart slide show?
5 Where is the retry logic in our code?	**6** Summarize this 3-page brief

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CHECK FOR UNDERSTANDING

Can every learner answer these?

1

Roughly how much text is one token?

2

Why do long chats cost more each turn?

3

What shares the context window?

4

What does caching make cheaper?

5

Name three ways AI fetches data.

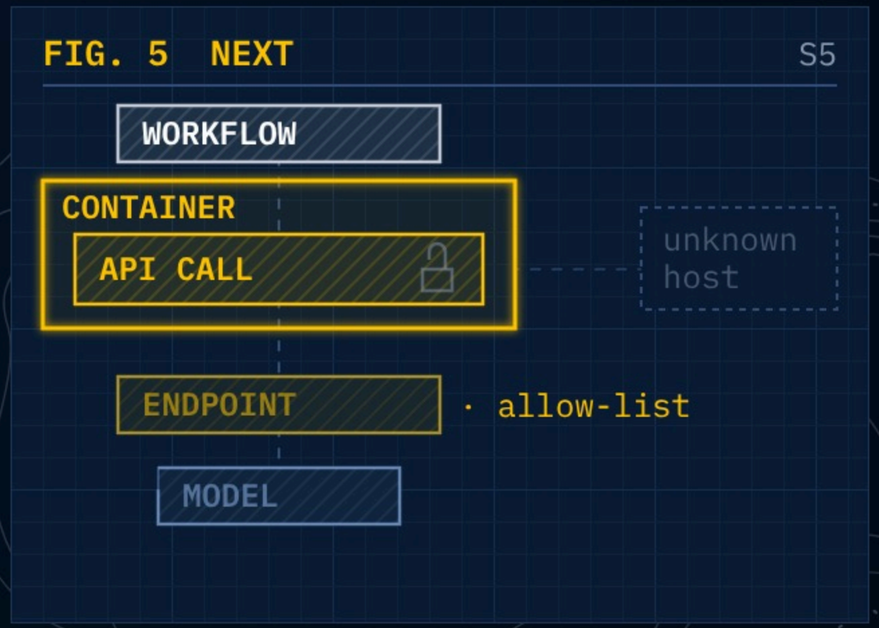
6

Why is a search index a security question?

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NEXT · SESSION 5 · APIS, ENDPOINTS AND CONTAINERS

How does your own software reach the model safely?



What Session 5 covers

- API calls
- Endpoints and config
- Containers
- Secrets and logging